AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Cardiovascular health

Cardiovascular disease (CVD) covers conditions of the heart and blood vessels, most of which begin with the slow build-up of fatty plaques in the arteries. This page explains how atherosclerosis and clotting lead to CVD, which factors raise the risk, how to read and design studies of risk, how obesity and cholesterol are measured, and how drugs treat the disease.

SPEC 1.10 Atherosclerosis

WHAT THIS SPEC POINT ASKS

Understand the sequence of events that leads to atherosclerosis: endothelial dysfunction, inflammatory response, plaque formation, raised blood pressure.

Learn this as a numbered chain. Six-mark answers are marked on the sequence and the links:

1. **Endothelial damage (dysfunction)**: the smooth inner lining of an **artery** is damaged, e.g. by **high blood pressure**, **chemicals in cigarette smoke**, or high blood LDL.
2. **Inflammatory response**: **white blood cells** (macrophages) move into the artery wall at the damaged site.
3. **Cholesterol** (from LDLs in the blood) builds up in the wall and is taken up by the white cells, forming **fatty streaks**.
4. Over time **calcium salts** and **fibrous tissue** build up too → an **atheroma (plaque)** forms. The artery wall hardens and becomes **less elastic**.
5. The plaque **narrows the lumen** → **raised blood pressure** → more damage to the endothelium → more plaque. This is **positive feedback**. A plaque can also **rupture**, which triggers a **blood clot** (1.11) that may block the artery completely.
6. When this happens in a **coronary artery**, less blood, so less O_2, reaches the **heart muscle cells** → less **aerobic respiration** → less ATP → the muscle can't contract properly and cells may die (a heart attack, 1.11).

Atherosclerosis happens in **arteries** (high pressure), not in veins or capillaries.

From endothelial damage to a blocked artery

1. The endothelium is damaged, for example by high blood pressure or by smoke chemicals.

2. The damaged lining lets LDL cholesterol enter the artery wall.

   Damaged endothelium is more permeable, so LDL particles pass into the wall more easily.

3. Inflammation attracts white blood cells, which take up the cholesterol and become foam cells.

4. Foam cells and fibrous material build a plaque, which narrows the lumen.

   Narrowing raises the resistance to blood flow, so blood pressure rises further.

5. The plaque may rupture and trigger a clot, which can completely block the artery.

   Blood flow past the blockage stops, so the tissue beyond it is deprived of oxygen.

EXAM TIP

Put the steps in order and link each to the next ("this causes…", "leading to…"). Always include **white blood cells / inflammation**, **cholesterol**, **plaque/atheroma** and **narrowing of the lumen**, and finish the chain: **coronary artery** → less O_2 to **heart muscle** → less **aerobic respiration**.

MARK-SCHEME WORDING

- "damage to the endothelial lining" (from high blood pressure, smoking, free radicals) → "inflammatory response" / white blood cells build up → "build-up of cholesterol" in the artery wall → "atheroma / plaque" → "narrowing / blocking" of the **coronary artery** → less "blood / oxygen to the heart muscle cells" for **aerobic** respiration.
- Say "plaque" or "atheroma", not "fatty deposits" or "hardening". Plaque is **not** "caused by blood clotting".
- CVD = any disease of the heart and blood vessels; CHD = disease of the coronary arteries.
- Stents "widen the lumen of the coronary artery", so more blood reaches the heart muscle for respiration.

Often asked as: explain how a factor (smoking, high blood pressure, saturated fat) leads to CVD, 6-mark chains with data, define CVD v CHD.

COMMON MISTAKE

Stopping halfway. Examiners insist on the words **coronary artery** and **heart muscle cells** short of **oxygen** for **aerobic** respiration. Bare "arteries", or "nutrients" instead of O_2, doesn't score.

In "deduce" questions, give conclusions from the data, not a description of it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Aneurysm: an artery wall weakened by atheroma bulges and may burst. **Angina**: chest pain when narrowed coronary arteries can't supply enough O_2 to the heart muscle during exertion.

Heart muscle short of O_2: cells switch to anaerobic respiration → lactic acid → lower pH denatures enzymes; much less ATP, so active transport and contraction fail (WBI11 Jan 2021).

SPEC 1.11 Blood clotting

WHAT THIS SPEC POINT ASKS

Understand the blood-clotting process (thromboplastin release, prothrombin → thrombin, fibrinogen → fibrin) and its role in cardiovascular disease (CVD).

The cascade

1. Damage to a blood vessel (or a ruptured plaque) exposes the tissue beneath; **platelets** stick to it and to each other.
2. Platelets and damaged tissue release **thromboplastin** (an enzyme).
3. Thromboplastin, with **calcium ions (Ca^{2+})**, catalyses the conversion of soluble **prothrombin** into **thrombin** (also an enzyme).
4. Thrombin catalyses the conversion of **soluble fibrinogen** into **insoluble fibrin**.
5. **Fibrin** forms a **mesh** that traps platelets and red blood cells → a **clot**.

Each product activates the next step, so a small trigger is amplified into a large response.

The link to CVD

- A ruptured atheroma triggers clotting **inside an artery**. The clot (**thrombus**) can block that artery, or a piece can break off and block a narrower vessel further on.
- Blocked **coronary artery** → heart muscle gets less O_2 → less aerobic respiration → less ATP → muscle cells die: a **myocardial infarction** (heart attack).
- Blocked artery to the **brain** → **stroke**.
- **Deep-vein thrombosis (DVT)**: a clot in a deep vein, usually in the leg (risk raised by inactivity, e.g. long flights, and by age).

MARK-SCHEME WORDING

- "platelets release thromboplastin". Platelets are the part most often left out.

- "thromboplastin is an enzyme" that "converts prothrombin into thrombin"; "thrombin is an enzyme" whose active site binds fibrinogen and converts it into fibrin.
- Fibrin forms a "mesh" that "traps platelets / blood cells" → stops bleeding and the entry of bacteria.
- Prothrombin is always present in the plasma as an **inactive precursor**.
- Soluble: thromboplastin, prothrombin, thrombin, fibrinogen. Insoluble: **fibrin** only. Active enzymes: thromboplastin and thrombin.
- Inhibitor questions: say what **won't** happen: "thrombin can't bind to fibrinogen", "less fibrinogen converted into fibrin", "less mesh to trap cells".
- Ca^{2+} and vitamin K are ignored in these answers: fine to mention, but they earn nothing.

Often asked as: cloze or table on the clotting factors, explain the effect of an inhibitor in a new context, 6-mark on DVT leading to pulmonary embolism.

COMMON MISTAKE

Saying fibrinogen turns into "fibres". It becomes **fibrin**, an **insoluble** protein. Remember **thrombin is a soluble enzyme**: it isn't part of the clot. Don't swap the solubility of thrombin and fibrin.

Retelling the whole cascade when the question asks about one step, or describing normal clotting when it asks what an **inhibitor stops**: that scores 1 mark at most.

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- **DVT → pulmonary embolism**: a clot from a leg vein travels vena cava → right atrium → tricuspid valve → right ventricle → pulmonary artery, and lodges where the arteries narrow (WBI11 Jan 2026).
- **Thrombolytic drugs**: plasmin hydrolyses the peptide bonds in fibrin, dissolving a clot (WBI11 Jun 2019).
- Thromboplastin is stored in platelets, **separate from prothrombin**, so clotting can't start until it's needed, then it's released quickly (WBI11 Jun 2022).

SPEC 1.12 Risk factors for CVD

WHAT THIS SPEC POINT ASKS

Know how genetics, diet, age, gender, high blood pressure, smoking and inactivity increase the risk of cardiovascular disease.

A **risk factor** increases the **probability** of a disease; it doesn't make it certain. Risk factors **add up**: someone who smokes, is inactive and eats a high-saturated-fat diet carries all three risks at once.

Risk factor	How it raises the risk of CVD	Can you change it?
Genetics	inherited alleles can cause high blood cholesterol (e.g. faulty LDL receptors), high blood pressure, or weaker arteries	no
Age	arteries lose elasticity and plaque builds up over a lifetime	no
Gender	men are at higher risk at younger ages; **oestrogen** in women before menopause raises HDL (protective)	no
High blood pressure	damages the endothelium → starts atherosclerosis; the heart works harder	partly (diet, exercise, drugs)
Diet	high saturated fat → high **LDL**/blood cholesterol → plaque; high **salt** → water retained → raised blood pressure; excess energy → **obesity**	yes
Smoking	chemicals **damage the endothelium**; **nicotine** triggers adrenaline release → **arterioles constrict** → raised heart rate and **blood pressure**; nicotine makes **platelets sticky** → clots; **carbon monoxide** binds Hb, so less O_2 reaches heart and brain → heart works harder; fewer **antioxidants**; lower **HDL**	yes
Inactivity	linked to obesity, high blood pressure and high LDL; regular exercise raises HDL and lowers blood pressure	yes

How smoking raises the risk of CVD

1. Carbon monoxide forms carboxyhaemoglobin, which carries less oxygen.

   Less oxygen reaches the heart and brain, so they work harder to compensate.

2. Nicotine releases adrenaline, which raises heart rate and constricts arteries.

   Constricted arteries have higher resistance, so blood pressure rises.

3. Smoking lowers HDL and increases free radicals.

   Less HDL means less cholesterol is removed from the artery walls, and free radicals damage the endothelium.

4. Platelets become more active, so clots form more readily.

   A clot on a damaged plaque can block the artery and cause a heart attack or stroke.

EXAM TIP

Explain the **mechanism**, not just "it's bad for you": link each factor to one of endothelial damage, raised blood pressure, raised LDL or increased clotting.

MARK-SCHEME WORDING

- Risk factor: "a variable that increases the chance of an event (the disease) happening".
- Non-lifestyle: age, sex, genetics. Lifestyle: smoking, BMI, salt, alcohol, inactivity. Always give the **direction**: "high BMI", "high salt intake". Bare "diet", "lifestyle" or "exercise" is ignored.
- Mechanisms that score: age → arteries "lose elasticity"; smoking → raises blood pressure and damages the endothelium; salt → raises blood pressure; high blood pressure → damages the endothelium.
- Describing a graph: "as the number of cigarettes smoked increases, the risk of CHD increases", "non-linear", there is still some risk with no smoking. Not "exponential".
- A study must adjust for confounders such as age and smoking, or it is "not valid". Reliability is ignored here.

Often asked as: classify factors as lifestyle or not, describe a risk graph, use a risk calculator, 6-mark comparing two risk calculators.

COMMON MISTAKE

Listing factors with no mechanism: that keeps a 6-mark answer at Level 1. "Poor diet" is too vague: say which part (high saturated fat, high salt) and how it acts.

Mixing up lifestyle and non-lifestyle factors. Two marks means two different points.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Risk calculators** (e.g. the Framingham score): add up the points for each factor and read off the % risk. Weaknesses: they leave out some factors, and people may guess their blood pressure or cholesterol (WBI11 Jun 2022).
- Women are partly protected **until the menopause** (WBI11 Jun 2023).

SPEC 1.13 Dietary antioxidants and CVD

WHAT THIS SPEC POINT ASKS

Understand the link between dietary antioxidants and the risk of CVD.

The idea

- Metabolism (and smoking) produces **free radicals**: unstable, highly reactive particles with an unpaired electron.
- Free radicals can **damage the endothelium** of arteries and **oxidise LDL**, making it more likely to be taken up into plaques. Both promote atherosclerosis.
- **Antioxidants** (e.g. **vitamins C and E**, polyphenols such as **flavonoids** in fruit, vegetables, tea and dark chocolate) react with and **neutralise free radicals**, so in theory they **reduce the risk** of CVD. Vitamin C is also needed to make collagen in artery walls.

The evidence is mixed

- **For**: many observational studies link high fruit and vegetable intake, or high blood vitamin C, with lower heart-attack rates.
- **Against**: large randomised trials and meta-analyses of **antioxidant supplements** found no clear benefit, and some suggest possible harm.
- People who eat lots of fruit and vegetables often also exercise more, smoke less and eat less saturated fat, so these **confounding variables** may explain the link. Correlation ≠ causation.

EXAM TIP

For "evaluate the evidence" questions: give the proposed mechanism, the supporting evidence, the conflicting evidence, then a balanced conclusion ("the evidence is inconclusive because…").

MARK-SCHEME WORDING

Learn this chain word for word. The same mark scheme keeps coming back:

- "antioxidants reduce free radicals" (neutralise / stabilise / donate electrons to them)
- → less "damage to the endothelial lining" (less oxidative stress)
- → "reduced plaque / atheroma formation"
- → lower risk of CVD.
- Question about *less* antioxidant? Flip it: "fewer free radicals are reduced", so the damage isn't prevented.
- Why eat a food that isn't proven to work? It's "not harmful" and is a food source anyway. "It contains antioxidants" scores 0.

Often asked as: explain how antioxidants lower CVD risk, compare the antioxidant content of foods, design a study (6-mark).

COMMON MISTAKE

Stopping before plaque and CVD. "Reduces CVD" on its own just repeats the question. Don't drift into LDL: this chain is about free radicals and endothelial damage.

SPEC 1.14 Core Practical 2: vitamin C content of food and drink

WHAT THIS SPEC POINT ASKS

Investigate the vitamin C content of food and drink.

Principle

- **DCPIP** is a **blue** dye. **Vitamin C** (ascorbic acid) is a **reducing agent** that turns DCPIP **colourless** (reduces it).
- The more vitamin C in a solution, the **less** of it you need to decolourise a fixed volume of DCPIP.

CORE PRACTICAL

Core practical 2: method in brief

1. Put a fixed volume (e.g. **1 cm³**) of DCPIP in a test tube.
2. Add a vitamin C solution of **known concentration** drop by drop from a burette or graduated pipette, **shaking gently** after each drop.
3. Stop when the blue colour **just disappears** (the end point); record the volume added.
4. Repeat (at least three times) and calculate a **mean**.
5. Repeat steps 1–4 with the fruit juice or food extract.
6. Optionally, test a range of known concentrations and plot a **calibration curve** of volume needed against concentration; read the unknown off the curve.

WORKED EXAMPLE

Worked example

1 cm³ of DCPIP is decolourised by **2.0 cm³** of a **1.0 mg cm^{-3}** vitamin C standard, so it takes **2.0 mg** of vitamin C.

The same 1 cm³ of DCPIP needs **8.0 cm³** of orange juice.

So 8.0 cm³ of juice contains 2.0 mg vitamin C → **0.25 mg cm^{-3}** of juice.

General rule: concentration of sample = (concentration of standard × volume of standard) ÷ volume of sample.

Variables and sources of error

- **Control**: volume and concentration of DCPIP, shaking, temperature, how the end point is judged (against a white tile).

- **Problems**: the end point is subjective; coloured juices mask the colour change (dilute them or use a colorimeter); vitamin C is **oxidised** by air and heat, so use fresh solutions and don't boil them.
- Safety: DCPIP is an irritant, so wear eye protection.

EXAM TIP

This practical hasn't been examined directly in any WBI11 paper from 2019 to 2026, but it's a named core practical, so still learn it. If it comes up, use the study-design language from 1.16: controlled variables make it **valid**; repeats and a mean make it **repeatable**.

SPEC 1.15 Interpreting data on illness and mortality

WHAT THIS SPEC POINT ASKS

Analyse and interpret quantitative data on illness and mortality to work out health risks, distinguishing correlation from causation and recognising conflicting evidence.

Key terms

- **Incidence**: number of **new** cases in a population over a set time (e.g. per 100 000 per year).
- **Prevalence**: number of people **with** the disease at a given time.
- **Mortality**: number of **deaths** from the disease (per 100 000 per year).
- **Relative risk**: risk in an exposed group ÷ risk in an unexposed group. A relative risk of 2.0 means twice as likely; 1.0 means no difference.

A three-step scaffold for data questions

1. **Describe**: state the trend, and **quote figures** from the data, with units (e.g. "relative risk rises from 1.0 to 2.4 between ages 40 and 70").
2. **Conclude**: say there is a **correlation** (a positive or negative association), only within the range of the data. Don't extrapolate to other ages, sexes or species.
3. **Evaluate**: is the link **causal**? Consider sample size, how the sample was chosen, controls, confounding variables, repeats, whether **error bars/standard deviations overlap** (non-overlapping bars suggest a **significant** difference), and whether a statistical test was done.

WORKED EXAMPLE

Worked example: percentage change

% change = (new − old) ÷ old × 100. Subtract first, then divide by the **old** value.

Deaths fall from 250 to 180 per 100 000: (180 − 250) ÷ 250 × 100 = **−28%**, a 28% decrease.

Give the d.p. or s.f. asked for, never a recurring decimal, and round numbers of people to whole numbers.

Correlation vs causation

- A **correlation** means two variables change together. It doesn't prove one causes the other: a **third (confounding) variable** might cause both.
- **Causation** is supported when there's a plausible **mechanism**, the relationship holds across many large, well-controlled studies, and changing the factor (e.g. in a trial) changes the outcome.
- **Conflicting evidence**: different studies may disagree because of different sample sizes or populations, uncontrolled variables, or bias. One study isn't enough; combine studies (meta-analysis) and look at the quality of each.

EXAM TIP

Always **use numbers from the data**, and use **correlation** or **association**, not "causes", unless the question gives you evidence of a mechanism or a controlled trial.

MARK-SCHEME WORDING

- Correlation: "a change in one variable is reflected by a change in another variable" / "a relationship between two variables". "Causes", "affects", "results in" and "leads to" are rejected. Add "but this doesn't show that X causes Y" where it fits.
- Error bars / SD: not overlapping → "the difference is (probably) **significant**"; overlapping → "probably not significant"; none shown → "no error bars / SD, so significance can't be judged". Link error bars to **significance**, never to "reliability".
- Evidence *against* a link: people with the disease at low exposure, or high exposure without the disease.
- A hazard ratio or relative risk of 1.00 is the baseline: a "means of comparison". With hazard ratios, say "risk of death", not risk of the disease.
- Predicting beyond the data: use the word "**extrapolate**".
- Why figures are only estimates: undiagnosed people, unregistered births and deaths, people who are hard to reach or count.

Often asked as: define correlation, state two conclusions from a graph, give evidence for and against a link, comment on data with error bars, calculate a percentage change.

COMMON MISTAKE

Narrating bar by bar instead of giving the trend. Giving one conclusion and its reverse as "two conclusions". Describing only the rise when the data also fall. Misquoting values: read them carefully and give units.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Statistical tests**: a t-test (or Mann-Whitney U) checks whether two groups differ significantly; a chi-squared test compares observed and expected numbers (WBI11 Oct 2021).
- Pooled data give a useful overview but can hide differences between groups (WBI11 Jan 2025).

SPEC 1.16 Designing studies of health risk factors

WHAT THIS SPEC POINT ASKS

Evaluate the design of studies used to find health risk factors, including sample selection and sample size, so the data are valid and reliable.

What makes a good study

- **Large sample**: reduces the effect of chance and of anomalous individuals → results are more **repeatable**, and statistics can be done.
- **Representative, random sample**: matches the population you want to draw conclusions about (age, sex, ethnicity, lifestyle). Random selection avoids **selection bias**.
- **Control group / matched groups**: compare people with and without the risk factor who are otherwise similar (**matched** for age, sex, diet, etc.).
- **Control confounding variables**: other factors that affect the outcome (smoking, diet, exercise, age). Controlling them makes the study **valid**.
- **Exclude** people who already have the disease: the study is about its **development**.
- **Long enough** duration for the disease to develop.
- **Repeatable and reproducible**: the method is published so others can repeat it; results agree.
- **Avoid bias**: **double-blind** trials (neither participants nor researchers know who gets what), a **placebo** in drug trials, independent funding, peer review.
- Measurement: self-reported data (e.g. diet questionnaires) are unreliable, because people forget, lie or are embarrassed.

Types of study

Study	How it works	Strengths / weaknesses
Cohort	follow a large group over time; record exposure and who develops the disease	can show which came first; takes years and is expensive; people drop out

Study	How it works	Strengths / weaknesses
Case-control	compare people who already **have** the disease with similar people who **don't**, and look back at past exposure	quick and cheap; relies on memory (recall bias)
Randomised controlled trial	randomly assign people to treatment or placebo	best evidence for causation; not ethical for harmful factors like smoking

Use the mark schemes' words:

- **Valid** = the study measures what it claims to, because only the factor being studied differs. Controls and matched groups give validity.
- **Repeatable** = you get similar results when you repeat it; **reproducible** = others get similar results. A large sample and repeats give this.
- Avoid the bare words "reliable", "accurate" and "precise": say which of valid, repeatable or reproducible you mean.

EXAM TIP

"Evaluate the design" means **strengths and weaknesses**: comment on the sample (size, how chosen, who's included), controls, confounding variables and duration, and say *why* each matters.

MARK-SCHEME WORDING

- Sample: "large groups" for "repeatability / reproducibility" and so statistics can be done. Participants "healthy / no known heart condition", "matched for age, sex, lifestyle". Side-effect trials use "healthy volunteers", not "patients".
- "control / placebo group"; "double blind … to avoid bias". In an animal injection study, the control gets **the liquid the drug was dissolved in**.
- Control the named variables "for validity"; if they aren't controlled, "the conclusions would not be valid".
- Duration: monitor the **incidence** of the disease over a long period (years for CVD, "because CVD develops slowly"); monitor side effects.
- Other criticisms: no control group, no statistics or error bars, no definition of "high" or "low" intake, "difficult to control all the variables".
- Say dose, volume or concentration, never "amount".

Often asked as: criticise a study (6-mark), design a study (6-mark), explain why a control or an exclusion was used, say whether a study is valid.

COMMON MISTAKE

Giving design points with no reason: each one needs a "so that…" (valid or repeatable). Inventing flaws the data don't show; "the groups must be equal sizes" isn't a valid criticism. Monitoring "risk" instead of **incidence**.

When asked whether a study is valid, **say which**, then give reasons.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Add a point specific to the context, e.g. cigarettes per day, ex-smokers, self-diagnosed v clinically diagnosed. That's what lifts a 6-mark criticism to full marks (WBI11 Jun 2021).
- A small sample can be justified for a very rare disease (WBI11 Oct 2021).

SPEC 1.17 Perceived risk vs actual risk

WHAT THIS SPEC POINT ASKS

Understand why people's perception of risk often differs from the actual risk, including underestimating or overestimating risks from diet and other lifestyle factors in heart disease.

- **Actual risk** is the measured probability of harm, based on scientific data (e.g. "1 in 100 per year").
- **Perceived risk** is how risky people *feel* something is.

People **overestimate** risk when…	People **underestimate** risk when…
it's unfamiliar, dramatic or reported in alarming media stories	it's familiar, everyday and **enjoyable** (e.g. fast food, smoking)
harm is sudden and severe (e.g. plane crash)	harm is **delayed** by years (heart disease develops slowly)
they know someone affected	they have no personal experience of it
it's imposed on them	they feel in control ("I'll cut down later")

Applied to heart disease: many people underestimate the risk from a high-saturated-fat or high-salt diet, inactivity, obesity and smoking, because the effects are slow, invisible and linked to enjoyable habits. They may overestimate small risks reported dramatically in the media (e.g. a scare about one food) and ignore large, steady ones. People also misunderstand statistics, e.g. "doubles the risk" sounds huge even when the starting risk is tiny.

MARK-SCHEME WORDING

- Underestimating: people "underestimate their mass / how much they smoke", "may not realise how much of a risk factor smoking can be". Self-reported data are unreliable.
- Risk calculators: they leave out other risk factors, and blood pressure or cholesterol values may be guessed.
- Consequences: overestimating → unnecessary lifestyle changes and stress; underestimating → no lifestyle change, testing or medication.
- A fair perception study uses people who don't know their own risk and aren't on CVD medication.

Often asked as: explain why someone misjudges their risk, compare perceived and actual risk on a bar chart, give the consequences of misjudging it.

COMMON MISTAKE

Describing a perceived-v-actual chart bar by bar. Give the overall pattern (e.g. perceived risk is generally lower than actual risk), then the exception.

SPEC 1.18 Cholesterol, HDL and LDL

WHAT THIS SPEC POINT ASKS

(i) Analyse data on how blood cholesterol, HDL and LDL levels might affect health. (ii) Know the evidence for a causal link between blood cholesterol (total and LDL) and CVD.

What they are

- **Cholesterol** is a lipid needed for **cell membranes** and to make steroid hormones and bile. It's insoluble, so it's carried in the blood as **lipoproteins** (lipid + protein).

	HDL (high-density lipoprotein)	**LDL** (low-density lipoprotein)
Made of	more protein; unsaturated fat + cholesterol	less protein; **saturated** fat + cholesterol
Carries cholesterol	**from tissues to the liver** (for removal or recycling)	**from the liver to body cells** (binds LDL receptors)
Effect	**lowers** blood cholesterol; may remove it from plaques	when receptors are saturated, LDL stays in the blood and is deposited in artery walls → **atheroma**

	HDL (high-density lipoprotein)	**LDL** (low-density lipoprotein)
Nickname	"good"	"bad"

- Diets high in **saturated fat** raise LDL; **unsaturated** fats raise HDL and lower LDL.
- The **ratio** of LDL to HDL (or total cholesterol to HDL) predicts risk better than either one alone: high LDL with low HDL is the worst combination.

Evidence that high cholesterol *causes* CVD

- A clear **mechanism**: LDL cholesterol accumulates in damaged artery walls and forms plaques (1.10).
- **Large studies** in many populations show a strong positive correlation between total/LDL cholesterol and CVD.
- **Intervention trials**: lowering LDL with **statins** (in randomised, placebo-controlled trials) reduces heart attacks and deaths; the bigger the fall in LDL, the bigger the fall in risk.
- **Genetic evidence**: people with **familial hypercholesterolaemia** (an inherited fault in LDL receptors) have very high LDL from birth and get CVD young.

Reading the data

- Quote values; compare groups; look at **error bars**: if the standard deviation bars **overlap**, the difference may **not be significant**.
- Consider confounders: people with high LDL may also smoke, be overweight or inactive.

MARK-SCHEME WORDING

- "the ratio / balance of LDL : HDL determines the risk". Risk is highest with high LDL **and** low HDL.
- "HDL transports cholesterol to the liver"; "LDL forms the plaque".
- LDL structure: the protein and phospholipid heads on the outside are **hydrophilic**, so they interact with the plasma; the cholesterol is surrounded inside.
- Data (e.g. a 3-D bar chart): at every HDL level, risk rises as LDL rises; at every LDL level, risk falls as HDL rises. Quote the evidence, don't just restate the conclusion.

Often asked as: explain HDL/LDL data, 6-mark on why one measurement alone doesn't predict risk, LDL structure and transport.

COMMON MISTAKE

Writing that HDL "is good cholesterol". Cholesterol is the same molecule: HDL and LDL differ in the **lipoprotein** carrying it and **which direction** they take it.

Leaving out how LDL mixes with the plasma. Calling the LDL receptor a channel or an enzyme. Describing the data without explaining it scores 1.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **LDL receptors**: LDL binds a receptor with a complementary shape and is taken into the cell by **endocytosis**. In familial hypercholesterolaemia a mutation changes the receptor's tertiary structure, so LDL can't bind and stays in the blood (WBI11 Oct 2023).
- Statins work by blocking **HMG-CoA reductase**, the liver enzyme that makes cholesterol (WBI11 Oct 2024).
- **Fibre** reduces the absorption of cholesterol from the gut; plant sterols also lower it (WBI11 Jan 2026).

SPEC 1.19 Using diet, exercise and smoking knowledge to reduce CHD risk

WHAT THIS SPEC POINT ASKS

Understand how people use scientific knowledge about diet (including obesity indicators such as BMI and waist-to-hip ratio), exercise and smoking to reduce their risk of coronary heart disease.

Energy balance

- **Energy in** (food) > **energy out** (metabolism + activity) → the surplus is stored as fat → weight gain and **obesity**.
- Energy in < energy out → body stores used → weight loss.
- Obesity raises CHD risk: raised blood pressure, raised LDL, and a higher risk of type 2 diabetes.

Body mass changes when the energy taken in as food and the energy used by the body are out of balance.

Obesity indicators

- **BMI = mass (kg) ÷ height² (m²)**.

BMI	Category
below 18.5	underweight
18.5–24.9	healthy
25–29.9	overweight

BMI	Category
30 and over	obese

WORKED EXAMPLE

Worked example: BMI and waist-to-hip ratio

Mass 84 kg, height 1.75 m: BMI = 84 ÷ 1.75² = 84 ÷ 3.0625 = **27.4** → overweight.

Waist 96 cm, hips 102 cm: waist-to-hip ratio = 96 ÷ 102 = **0.94** (or 0.94 : 1). That's above about 0.90 (men) or 0.85 (women), so the risk is raised.

Rearranged: mass = BMI × height². A BMI of 30 at 1.75 m is 30 × 3.0625 = **91.9 kg**.

Height in cm? Convert to m **before** squaring.

Worked example: BMI for 95 kg and 1.70 m

1. Write the formula: BMI = mass ÷ height², with mass in kg and height in m.
2. Square the height: 1.70 × 1.70 = 2.89 m².
3. Divide the mass by the squared height: 95 ÷ 2.89 = 32.87.
4. Compare with the table: 32.87 is above 30, so the person is classed as obese.

- **Limitations of BMI**: doesn't distinguish fat from muscle (athletes can score "overweight"), and ignores **where** fat is stored.
- **Waist-to-hip ratio** measures **abdominal (central) fat**, which is more closely linked to CHD, so it can be a better predictor.
- Other obesity indicators: **waist circumference**, **waist-to-height ratio**, **skinfold thickness**, **% body fat**. Weight on its own isn't an indicator of obesity.

How people act on the science

- **Diet**: less saturated fat and salt, more fruit, vegetables and fibre; reading food labels (traffic-light labels for fat, saturated fat, sugar and salt); controlling energy intake.
- **Exercise**: regular aerobic activity lowers blood pressure and LDL, raises HDL and helps control weight.
- **Smoking**: stopping reduces risk quickly (support services, warnings on packets, advertising bans, smoke-free laws).

MARK-SCHEME WORDING

- Indicators other than BMI: "waist-to-hip ratio", "waist (circumference)", waist-to-height ratio, "skinfold thickness", % body fat. Weight and "pinch test" are ignored, and so is BMI when the question says "other than BMI".
- BMI = mass (kg) ÷ height² (m²). 25 to under 30 = overweight; 30 or more = obese. WHR = waist ÷ hip.
- Diet changes, each with its reason: less saturated fat / cholesterol → less atheroma; less salt → lower blood pressure, less endothelial damage; more antioxidants → reduce free radicals; more fibre → less cholesterol absorbed; fewer calories.
- Weight loss: "use more energy than is consumed", so eat less **and** exercise more. It must be a change: "regular exercise" alone is ignored.

Often asked as: calculate and classify a BMI, rearrange the BMI or WHR formula, name an indicator other than BMI, recommend diet changes with reasons.

COMMON MISTAKE

Giving a risk factor when an **indicator** was asked for, or giving only one. "Change your diet" is too vague: name the change and its direction.

Putting smoking in a "diet" answer. Giving the reason as just "increases the risk of CVD": say how (e.g. raises blood pressure).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Skinfold callipers** measure fat under the skin. Limits: fat isn't spread evenly, exercise can replace fat with muscle, and readings mislead in pregnancy (WBI11 Jun 2021).
- WHR measuring errors: the tape pulled too tight at the waist, or the person breathing in (WBI11 Jun 2021).

SPEC 1.20 Treatments for CVD: benefits and risks

WHAT THIS SPEC POINT ASKS

Know the benefits and risks of treatments for CVD: antihypertensives, statins, anticoagulants and platelet inhibitors.

For each drug, learn the chain: **what it does → what that prevents in 1.10/1.11 → its risks**.

Treatment	How it works	Benefit	Risks / side effects
Antihypertensives (e.g. ACE inhibitors, beta-blockers, diuretics, calcium-channel blockers)	lower blood pressure: widen blood vessels, slow the heart, or reduce blood volume	less endothelial damage → less atherosclerosis; lower risk of stroke and heart attack	dizziness, fainting, tiredness, palpitations, kidney problems; may need several drugs
Statins	block a liver enzyme used to make cholesterol → lower blood **LDL**	fewer plaques form → lower risk of heart attack and stroke	muscle aches and weakness, liver damage, digestive upsets; small increased risk of diabetes; lifelong use
Anticoagulants (e.g. warfarin, heparin)	interfere with clotting factors (e.g. less prothrombin made)	fewer clots → lower risk of thrombosis, stroke, DVT	**excessive bleeding**/haemorrhage when injured; dose must be monitored; interacts with other drugs and some foods
Platelet inhibitors (e.g. aspirin, clopidogrel)	stop platelets sticking together	fewer clots form at damaged arteries	**bleeding**, including stomach bleeding and ulcers; allergic reactions

Anticoagulants vs platelet inhibitors: both reduce clotting, but anticoagulants act on **clotting factors** in the plasma, while platelet inhibitors act on **platelets**.

General issues with all drug treatment: side effects, the cost of lifelong medication, and the fact that drugs treat the risk without changing the lifestyle causing it.

EXAM TIP

"Benefits and risks" questions are marked as pairs. For each drug give **one benefit linked to the disease process** and **one specific risk**. "Bleeding" alone is too vague: say *excessive bleeding if injured* or *internal/stomach bleeding*.

MARK-SCHEME WORDING

- Antihypertensives lower blood pressure; statins lower blood cholesterol (LDL); anticoagulants prevent clotting; platelet inhibitors reduce clotting.
- Anticoagulant risks: "bleeds too easily / internally", bruising, nosebleeds, heavy periods. Dose too high → "clotting time too long … too much blood could be lost"; too low → clots form → heart attack, stroke, embolism or DVT.

- Statin side effects: muscle pain, headache, nausea, liver damage, fatigue, diabetes. Learn 2–3 for each drug class.
- Why use several drugs: "many factors cause CVD", so "different drugs treat different conditions".

Often asked as: match a drug to its use (MCQ), name a risk or side effect, explain why a warfarin dose is monitored, calculate a percentage change in clotting time.

COMMON MISTAKE

Confusing a **side effect** (e.g. headache, bruising) with a **risk** (e.g. a serious bleed, or a clot if the dose is too low). Not knowing which drug treats which condition.

Calling warfarin a non-competitive inhibitor. Writing "gene" when you mean **allele** caps a pharmacogenomics answer at Level 1.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Warfarin** has a similar shape to vitamin K and **competitively** inhibits the enzyme that recycles it, so less prothrombin is made. Extra vitamin K competes with warfarin for the active site (WBI11 Oct 2020).
- **INR** (international normalised ratio) is used to monitor warfarin: too high → bleeding risk, too low → clotting risk, so it is kept within a safe range (WBI11 Oct 2022).
- **Pharmacogenomics**: people with alleles for slow warfarin breakdown need a lower dose (WBI11 Oct 2024).

Quick checks

QUICK CHECK

Which step is the earliest in the development of atherosclerosis?

- Formation of a blood clot on the plaque
- Narrowing of the lumen by fatty plaque
- Damage to the endothelium
- Rupture of an aneurysm

Atherosclerosis begins when the endothelial lining is damaged, which allows cholesterol and white blood cells into the wall. Clots and narrowing come later. Students often pick narrowing, because it is the most visible result, but it is not the first step.

QUICK CHECK

A person has a mass of 95 kg and a height of 1.70 m. What is their BMI?

- 55.8
- 25.9
- 32.9
- 0.56

BMI = mass ÷ height², so 95 ÷ (1.70 × 1.70) = 95 ÷ 2.89 = 32.87, which rounds to 32.9 and falls in the obese range. The wrong answers come from dividing by height rather than height squared, or by dividing in the wrong order.

QUICK CHECK

Why does smoking raise blood pressure?

- Carbon monoxide makes the blood thicker
- Nicotine makes the heart beat more slowly
- Nicotine releases adrenaline, which raises heart rate and constricts arteries
- Smoke causes the kidneys to hold water

Nicotine stimulates adrenaline secretion, increasing heart rate and narrowing arteries. Both raise blood pressure. Carbon monoxide reduces the oxygen the blood carries rather than making it thicker, which is a common confusion.

QUICK CHECK

Which drug type works by stopping the liver from making cholesterol?

- Beta blockers
- Diuretics
- Statins
- Aspirin

Statins inhibit HMG-CoA reductase, the enzyme that makes cholesterol in the liver. Beta blockers lower heart rate, diuretics remove water and salt, and aspirin reduces platelet stickiness. Those three work on different targets, which is why they are often confused.